Math 105L **Lecture 7-1**

Compound Interest & Developing Exponential Growth

Preliminaries: Investing at 6% Annual Interest

Suppose you invest an initial principal $P_0 = \$10$ at $r = 6\%$ (0.06) annual interest, compounded annually:

- **After 1 year ($t = 1$):**
$$\$10 + 0.06 \times \$10 = \$10(1 + 0.06) = \$\mathbf{10.60}$$
- **After 2 years ($t = 2$):**
$$10.60 + 0.06 \times 10.60 = 10.60(1 + 0.06) = 10(1 + 0.06)(1 + 0.06) = \mathbf{10(1.06)^2} = \$\mathbf{11.236}$$
- **After 10 years ($t = 10$): $\mathbf{10(1.06)^{10}} \approx \$\mathbf{17.91}$**
- **After t years: $\mathbf{P(t) = 10(1.06)^t}$** (Standard annual compounding: $P(t) = P_0(1 + r)^t$)

Varying the Compounding Frequency (n Periods Per Year)

Suppose instead you invest \$10 at 6% annual interest, compounded semi-annually ($n = 2$ times per year). That means each half-year you receive $\frac{6\%}{2} = 3\%$ interest:

- **After 1 year (2 periods):** $10\left(1 + \frac{0.06}{2}\right)^2 = 10(1.03)^2 = \$10.609 > \$10.60$.
- **After 2 years (4 periods):** $10\left(1 + \frac{0.06}{2}\right)^4 = 10(1.03)^4 \approx \$11.255$.
- **After 10 years (20 periods):** $10\left(1 + \frac{0.06}{2}\right)^{20} \approx \$18.06$.
- **General Compounding Formula (n periods per year):**
$$\mathbf{P(t) = P_0\left(1 + \frac{r}{n}\right)^{nt}}$$
- **Why does more frequent compounding yield more money?** Because interest earned in earlier periods is added to the balance and itself earns interest in subsequent periods (*interest on interest*).

Doubling the Principal vs. Doubling the Interest Rate

- **Option A (Double Principal):** \$20 at 6% annual interest $\implies P_A(t) = 20(1.06)^t$.
- **Option B (Double Rate):** \$10 at 12% annual interest $\implies P_B(t) = 10(1.12)^t$.
- **Short-Term Horizon ($t = 10$ years):**
$$P_A(10) = 20(1.06)^{10} \approx \$\mathbf{35.82}, \qquad P_B(10) = 10(1.12)^{10} \approx \$\mathbf{31.06} \quad \text{(Option A wins!)}$$
- **Long-Term Horizon ($t = 100$ years):**
$$P_A(100) = 20(1.06)^{100} \approx \$\mathbf{6,786}, \qquad P_B(100) = 10(1.12)^{100} \approx \$\mathbf{835,220} \quad \text{(Option B crushes Option A!)}$$
- **Core Pedagogical Insight:** A linear multiplier on the initial principal (P_0) only scales linearly, whereas an increase in the exponential base compounds multiplicatively across time.

Pushing the Limit: Continuous Compounding & The Number e

Pushing the Compounding Frequency to Infinity

Suppose we invest $P_0 = \$1$ at $r = 100\%$ ($r = 1$) interest for $t = 1$ year. How much do we have if compounding occurs:

- **Annually** ($n = 1$)**:** $1\left(1 + \frac{1}{1}\right)^1 = \$2.00$
- **Semiannually** ($n = 2$)**:** $1\left(1 + \frac{1}{2}\right)^2 = (1.5)^2 = \$2.25$
- **Monthly** ($n = 12$)**:** $1\left(1 + \frac{1}{12}\right)^{12} \approx \mathbf{\$2.6130}$
- **Daily** ($n = 365$)**:** $1\left(1 + \frac{1}{365}\right)^{365} \approx \mathbf{\$2.7145}$
- **Every Minute** ($n = 525,600$)**:** $1\left(1 + \frac{1}{525,600}\right)^{525,600} \approx \mathbf{\$2.718279}$
- **Every Second** ($n = 31,536,000$)**:** $1\left(1 + \frac{1}{31,536,000}\right)^{31,536,000} \approx \mathbf{\$2.71828178}$

Does the balance grow without bound? **No!** It converges to a precise finite mathematical limit:

Definition: Euler's Number e

The mathematical constant e is defined as the limit of continuous growth on unit capital at 100% interest over one unit of time:

$$\mathbf{e} = \lim_{\mathbf{n}\to\infty}\left(\mathbf{1} + \frac{\mathbf{1}}{\mathbf{n}}\right)^{\mathbf{n}} \approx \mathbf{2.718281828459}\ldots$$

General Continuous Compounding Formula

For an arbitrary annual interest rate r and principal P_0, we let $n \to \infty$ in the discrete compounding formula:

$$\lim_{n\to\infty} P_0\left(1 + \frac{r}{n}\right)^{nt} = P_0 \lim_{n\to\infty}\left[\left(1 + \frac{1}{n/r}\right)^{n/r}\right]^{rt}$$

Letting $m = \frac{n}{r}$, as $n \to \infty$, we have $m \to \infty$:

$$P_0\left[\lim_{m\to\infty}\left(1 + \frac{1}{m}\right)^{m}\right]^{rt} = \mathbf{P_0 e^{rt}}$$

The Continuous Growth Law:

When compounding occurs *continuously* ($n \to \infty$), the accumulated balance after t years is:

$$\mathbf{P(t) = P_0 e^{rt}}$$

Example from Class: Investing \$1 at 5% ($r = 0.05$) compounded continuously for 1 year yields:

$$P(1) = 1 \cdot e^{0.05 \cdot 1} = e^{0.05} \approx \mathbf{\$1.05127}$$